Semantic and Instance Segmentation

Per-pixel labels: FCN, U-Net, Dice, and Mask R-CNN

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From boxes to pixels

The computer-vision task ladder

Each task asks **more** of the same image.

classification	→	detection	→	segmentation
what?		**what + where (box)**		**what at each pixel**

last lecture ended at **detection**; today we go all the way to **every pixel**.

Recap: what detection gave us

For each object, detection returns a **box + class + score**:

$$x \;\to\; \left\{\left(\hat{b}_i, \hat{y}_i, \hat{s}_i\right)\right\}_{i=1}^{N}$$

- $\hat{b}_i$ — a **rectangle** (x, y, w, h);
- $\hat{y}_i$ — the class;
- $\hat{s}_i$ — a confidence, filtered by **NMS**.

A box says **“an object is roughly here.”** It never traces the object’s actual **shape**.

The limit of a box

- a box is **axis-aligned** — a diagonal or curved object wastes most of it;
- overlapping objects share pixels — which pixel belongs to whom?
- some questions need the **outline**: tumor area, road surface, free space.

we want a label for every pixel, not a rectangle

Segmentation: label every pixel

a toy 6 × 8 label map — one class per pixel

sky	sky	sky	sky	sky	sky	sky	sky
sky	sky	bldg	bldg	bldg	sky	sky	sky
bldg	bldg	bldg	bldg	bldg	bldg	bldg	bldg
road	road	road	road	road	road	road	road
road	car	car	road	road	car	car	road
road	road	road	road	road	road	road	road

the output has the **same spatial size** as the input — one class per pixel.

Assign a class to **every** pixel (h, w):

$$x \quad \rightarrow \quad y_{hw} \in \{1, ..., K\} \quad \forall\, h, w$$

- output is a **label map** of size $H \times W$;
- **all** pixels of the same class look identical — two cats merge into one “cat” region;

Semantic segmentation answers **“what class is this pixel?”** — it does **not** count objects.

Instance segmentation — the task

Return a **mask per object**, like detection but with pixels:

$$x \quad \rightarrow \quad \left\{ \left(\hat{y}_i, \hat{b}_i, \hat{m}_i, \hat{s}_i \right) \right\}_{i=1}^{N}$$

- $\hat{m}_i$ — a **binary mask** over the object's pixels;
- two cats now become **two separate** masks — objects are **counted and separated**.

detection's boxes, upgraded to pixel-accurate masks

Panoptic segmentation — one label for everything

Give **every** pixel a pair — a class **and** an instance id:

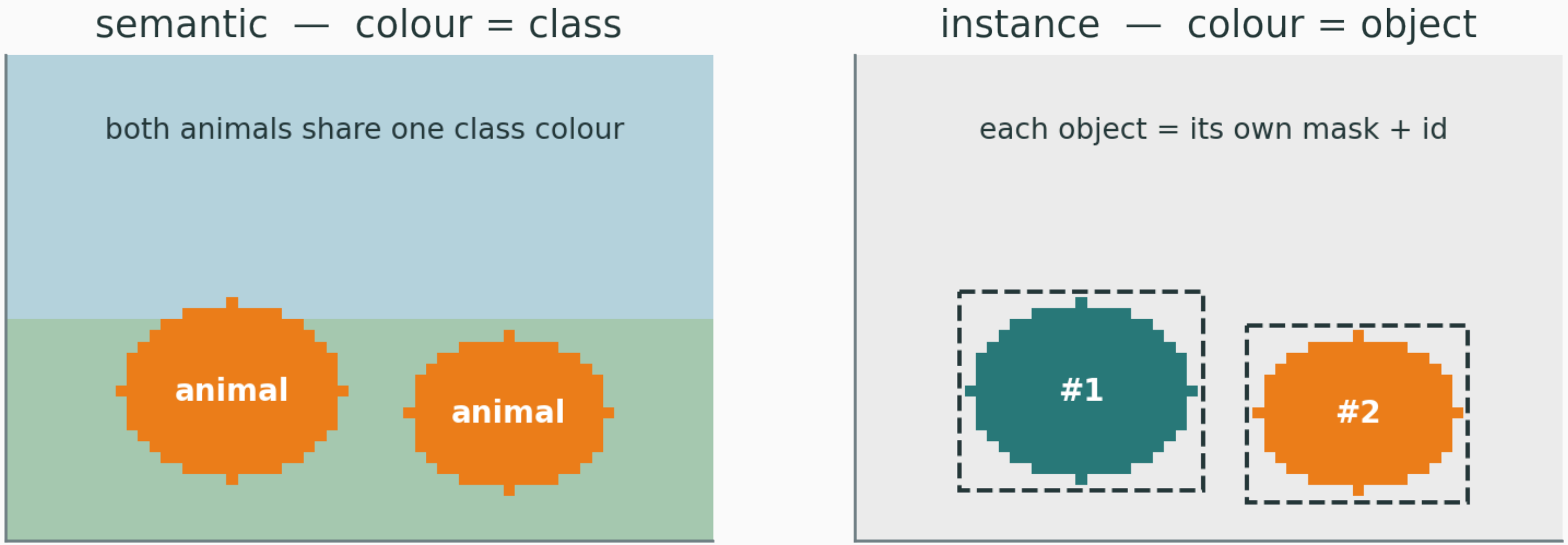
$$x \quad \to \quad (c_{hw}, \mathrm{id}_{hw}) \quad \forall\, h, w$$

stuff — amorphous regions (sky, road, grass): class only, **no instance**.

things — countable objects (car, person): class **and** a distinct instance id.

panoptic = semantic (stuff) **+** instance (things), with **no gaps or overlaps**.

Semantic vs instance, side by side



same scene · **left**: colour by class · **right**: colour by object.

Detection gives **boxes**;
segmentation labels every pixel.

Task	**Output**	**Counts objects?**
semantic seg	per-pixel class map	no
instance seg	per-object mask	yes
panoptic seg	per-pixel (class, id)	yes (things only)

Semantic segmentation

Per-pixel classification

Segmentation is **classification, run at every pixel**:

$$p_{hwk} = p(y_{hw} = k \mid x)$$

- for each pixel (h, w) the network emits a vector of K class scores;
- a **softmax over the K channels** turns scores into probabilities;

> Same loss you already know from image classification — just applied $H \times W$ times, once per pixel.

The output is a tensor

A classifier outputs **one** vector of length K. Segmentation outputs **a grid** of them:

$$\text{logits} \in \mathbb{R}^{H \times W \times K} \quad \rightarrow \quad \text{softmax over} \quad K \quad \rightarrow \quad p \in \mathbb{R}^{H \times W \times K}$$

the **channel** dimension holds class probabilities; argmax_k per pixel gives the label map.

Pixelwise cross-entropy

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Sum the ordinary cross-entropy over **all pixels** and average:

$$\mathcal{L} = -\frac{1}{HW} \sum_{h,w} \log p_{hw,\, y_{hw}}$$

- y_{hw} — the **true** class at pixel (h, w);
- $p_{hw,\, y_{hw}}$ — the probability the model gave to that **correct** class;

each pixel contributes its own classification loss

Reading the loss

$$\mathcal{L} = -\frac{1}{HW}\sum_{h,w} \underbrace{-\log p_{hw,\,y_{hw}}}_{\text{loss at one pixel}}$$

- a **confident, correct** pixel ($p \to 1$) contributes ≈ 0;
- a **confident, wrong** pixel ($p \to 0$) contributes a **large** penalty;

the $\frac{1}{HW}$ makes it a **mean** — it treats every pixel as equally important (we will question this soon).

We need dense prediction

- a classification CNN ends in **pooling + fully-connected** layers → one vector;
- that **destroys spatial layout** — exactly what a pixel map needs to keep.

how do we turn a "one-label" network into an "$H \times W$-label" network?

Fully convolutional networks (FCN)

Replace the final **fully-connected** layers with 1×1 **convolutions**:

- a 1×1 conv is an FC layer applied **at every spatial location**;
- the net now accepts **any input size** and emits a **coarse label map**, not a vector;

> "Fully convolutional" = **no FC layers at all**. Output is a spatial map of class scores — the first end-to-end dense-prediction net.

FCN: coarse semantics meet fine detail

Deep layers know **what**; shallow layers know **where**.

Deep features — rich **semantics**, but **coarse** (heavily downsampled).

Shallow features — precise **location / edges**, but weak semantics.

FCN **fuses** a coarse deep prediction with shallow appearance to sharpen boundaries.

Encoder–decoder

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The general shape of every modern segmentation net:

encoder shrinks space to gather context · **decoder** grows it back to pixel resolution.

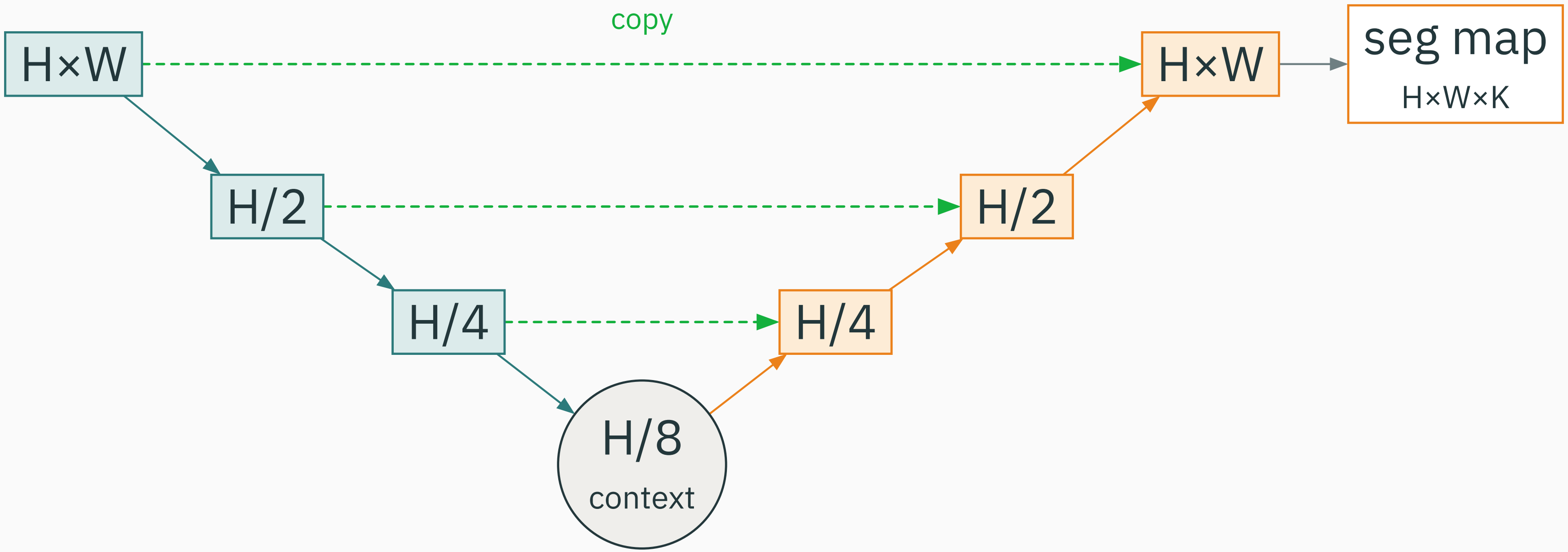
Upsampling in the decoder

The decoder must **increase** spatial resolution. Two common ways:

- **transposed convolution** — a learned upsampling ("deconv");
- **bilinear upsample + conv** — cheap interpolation, then refine.

> Downsampling was **pooling / stride**; upsampling **undoes** it — but the fine detail that pooling threw away is gone unless we bring it back another way.

Symmetric encoder–decoder with **skip connections** at every level:



contracting path (teal) captures context · expanding path (orange) localizes · green skips copy detail across.

Why skip connections

The bottleneck knows **what** but has lost **where**. Skips restore the **where**.

- each skip **copies** high-resolution encoder features to the matching decoder level;
- the decoder **concatenates** them with its upsampled features before predicting;

coarse semantics (from depth) + fine detail (from skips) = sharp masks

U-Net: born in medical imaging

optional

- designed for **biomedical** segmentation, where images are large and labels scarce;
- works from **very few** annotated images (heavy augmentation);
- the skip-connected encoder–decoder is now the **default backbone** for dense prediction — including diffusion models.

Interactive: encoder–decoder / U-Net

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↗ nipunbatra.github.io/interactive-articles/unet

Walk an image down the contracting path and back up the expanding path — toggle the skip connections and watch the predicted mask lose, then recover, its fine boundaries.

Q. If you **remove** every skip connection, what happens to the sharpness of the output mask, and why?

Losses for dense prediction

In dense prediction the **background usually dominates** the image.

- a small tumor may be **< 1%** of the pixels; the rest is "healthy";
- pixelwise CE is a **mean over pixels** → the majority class drowns the minority.

A network can score **low CE and high pixel accuracy** by predicting **"all background"** — and still be useless.

The Dice coefficient

Measure **overlap** between predicted set P and ground-truth set G:

$$\mathrm{Dice} = \frac{2|P \cap G|}{|P| + |G|}$$

- $= 1$ when P and G coincide exactly; $= 0$ when they are disjoint;
- it **ignores** the vast background — it only cares about the **foreground overlap**.

same idea as the F1 score, phrased for masks.

Make Dice differentiable: use **probabilities** $p_i \in [0,1]$ instead of hard sets, and **minimize** $1 - \mathrm{Dice}$:

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon}$$

- p_i — predicted foreground probability at pixel i; $g_i \in \{0,1\}$ — ground truth;
- ε — a small constant so an empty mask is **stable**, not $\frac{0}{0}$.

gradient pushes overlap up directly, regardless of how much background there is

Worked Dice example

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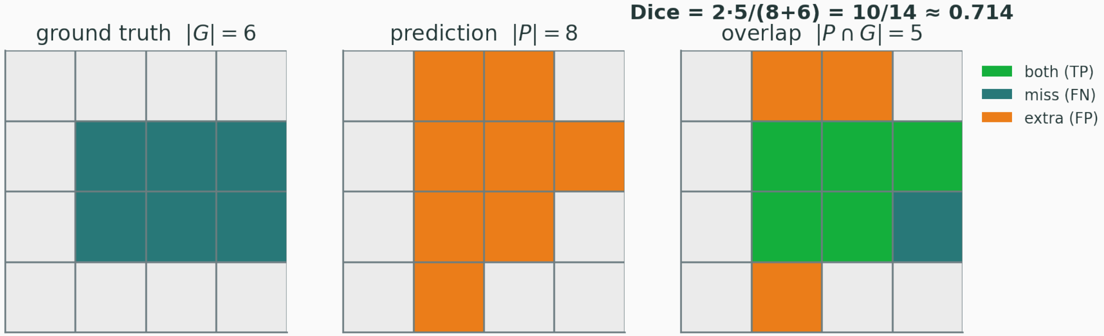


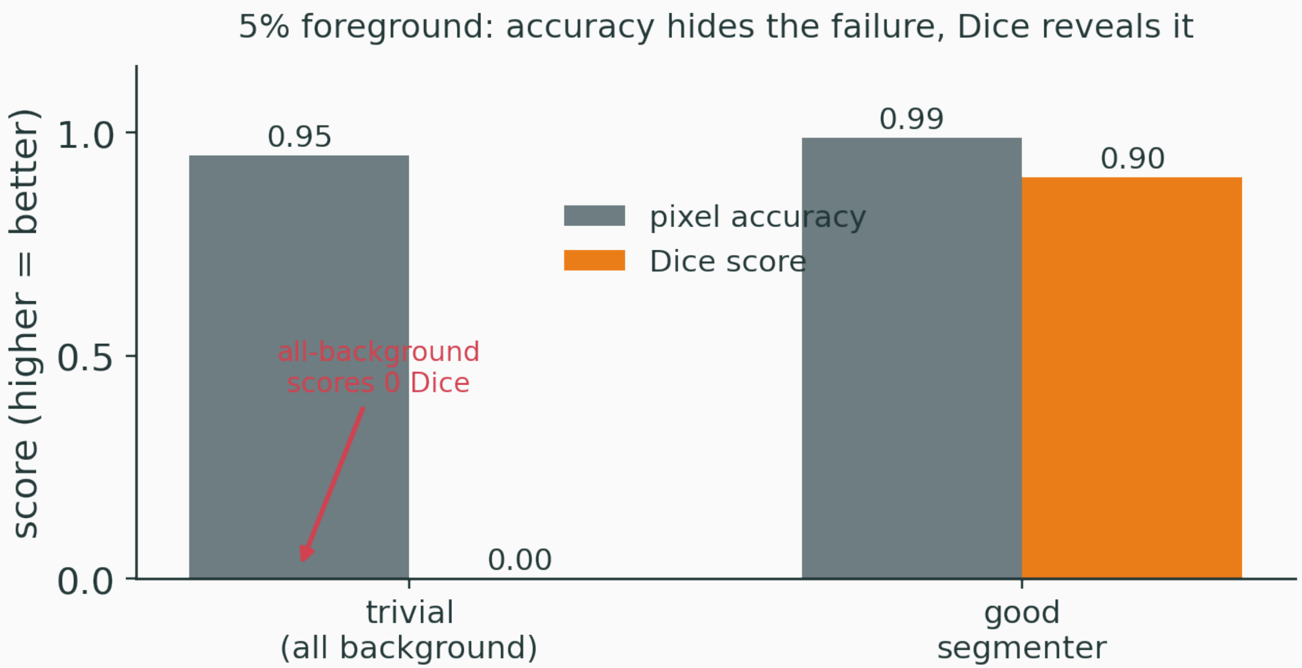
On a 4×4 grid:

- $|G| = 6$ (true region),
- $|P| = 8$ (predicted),
- $|P \cap G| = 5$ (overlap).

$$\text{Dice} = \frac{2 \cdot 5}{8 + 6} = \frac{10}{14} \approx 0.714$$

green = correct (TP) · teal = missed (FN) · orange = extra (FP).

Imbalanced scene (5% foreground), two "models":



pixel accuracy calls both **"great"**; Dice **exposes** the trivial all-background predictor.

CE vs Dice — use both

They fail in **opposite** ways, so practitioners **combine** them:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda\,\mathcal{L}_{\mathrm{Dice}}$$

Cross-entropy — smooth, stable gradients per pixel; but **imbalance-blind**.

Dice — directly targets **overlap**; but noisier gradients, unstable on empty masks.

CE gives a clean training signal; Dice keeps the **foreground** honest.

Interactive: segmentation losses

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↗ nipunbatra.github.io/interactive-articles/image-segmentation

Paint a prediction over a ground-truth mask and watch pixel CE and the Dice score move — see how a mostly-background prediction can keep CE low while Dice collapses to zero.

Q. Why does the trivial "predict all background" solution give **Dice** $= 0$ even though its pixel **accuracy** is 95%?

Instance segmentation

Semantic seg cannot **separate** two touching cats — both are just “cat” pixels.

instance seg = detect each object, then segment inside its box

so we bolt a **mask** onto a detector — one mask per detected object.

Recap: Faster R-CNN

The two-stage detector we build on:

$$\text{backbone} \to \text{RPN (proposals)} \to \text{RoI features} \to \{\text{class}, \text{box}\}$$

- the **RPN** proposes candidate regions;
- each region's features feed **two** heads: a **class** head and a **box-regression** head.

Faster R-CNN already localizes objects — it just stops at boxes.

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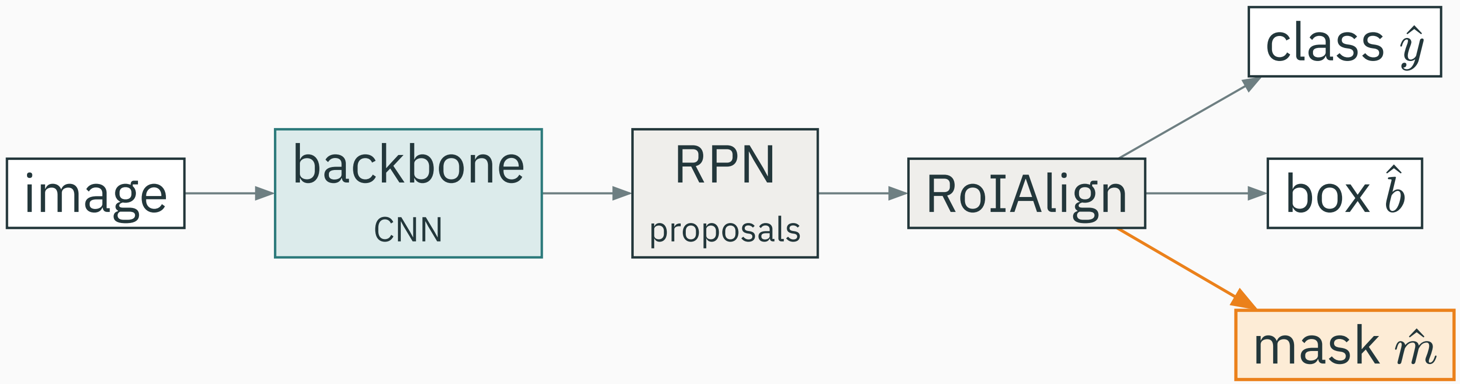
Add a **third, parallel** head that predicts a **mask**:



same backbone + RPN · three heads: **class**, **box**, and the new **mask** branch.

The mask branch

For each RoI, a **small FCN** predicts one binary mask per class:

- output is K masks of size $m \times m$ (e.g. 28×28), one per class;
- the **classification head chooses which** mask to use — the mask branch stays **class-agnostic** in its loss;

Decoupling **“what class”** from **“which pixels”** is the key trick: masks and classes do not compete.

Mask loss

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Per-pixel **binary** cross-entropy, only on the RoI's chosen-class mask:

$$\mathcal{L}_{\text{mask}} = -\frac{1}{m^2}\sum_{u,v}\left[\, m_{uv}\log \hat{m}_{uv} + (1 - m_{uv})\log(1 - \hat{m}_{uv})\,\right]$$

- $m_{uv} \in \{0, 1\}$ — true mask; $\hat{m}_{uv} \in [0, 1]$ — predicted (per-pixel **sigmoid**);
- summed only over the $m \times m$ RoI, and only for the **ground-truth class**.

total loss: $\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{mask}}$

RoIAlign — why alignment matters

The old **RoIPool rounded** region coordinates to the feature grid.

- rounding shifts features by up to **half a cell** — invisible for a coarse box;
- for a **per-pixel mask** that misalignment **smears the boundary**.

RoIAlign skips rounding: it samples features at **exact** sub-pixel locations with bilinear interpolation — the single change that made masks sharp.

Summary

Task → loss, at a glance

Task	Output	Loss
semantic seg	per-pixel class map	pixel CE (+ Dice)
instance seg	per-object mask	detection loss + mask BCE
panoptic seg	per-pixel (class, id)	both, fused

the **output shape** dictates the **loss**.

Segmentation methods

Method	**Core idea**	**Gives**
FCN	FC $\rightarrow 1 \times 1$ conv: dense conversion	coarse class map
U-Net	encoder–decoder + skip connections	sharp class map
Mask R-CNN	detector + parallel mask branch	per-object masks

FCN made dense prediction possible; U-Net made it **sharp**; Mask R-CNN made it **per-object**.

Where segmentation is used

optional

- **medical imaging** — tumor / organ delineation (U-Net's home turf);
- **autonomous driving** — drivable surface, lanes, pedestrians (panoptic);
- **photo editing** — background removal, object selection;
- **remote sensing** — land cover, buildings, roads from satellite images.

Final mental model — one idea

Classification asks **what?**

Detection asks **what + where (box)?**

Segmentation asks: what at each pixel?

coarser → finer: one label · a few boxes · a whole grid of labels

Detection gives boxes; segmentation labels every pixel.

From grids of pixels to sequences of tokens — next, **sequence models**.